

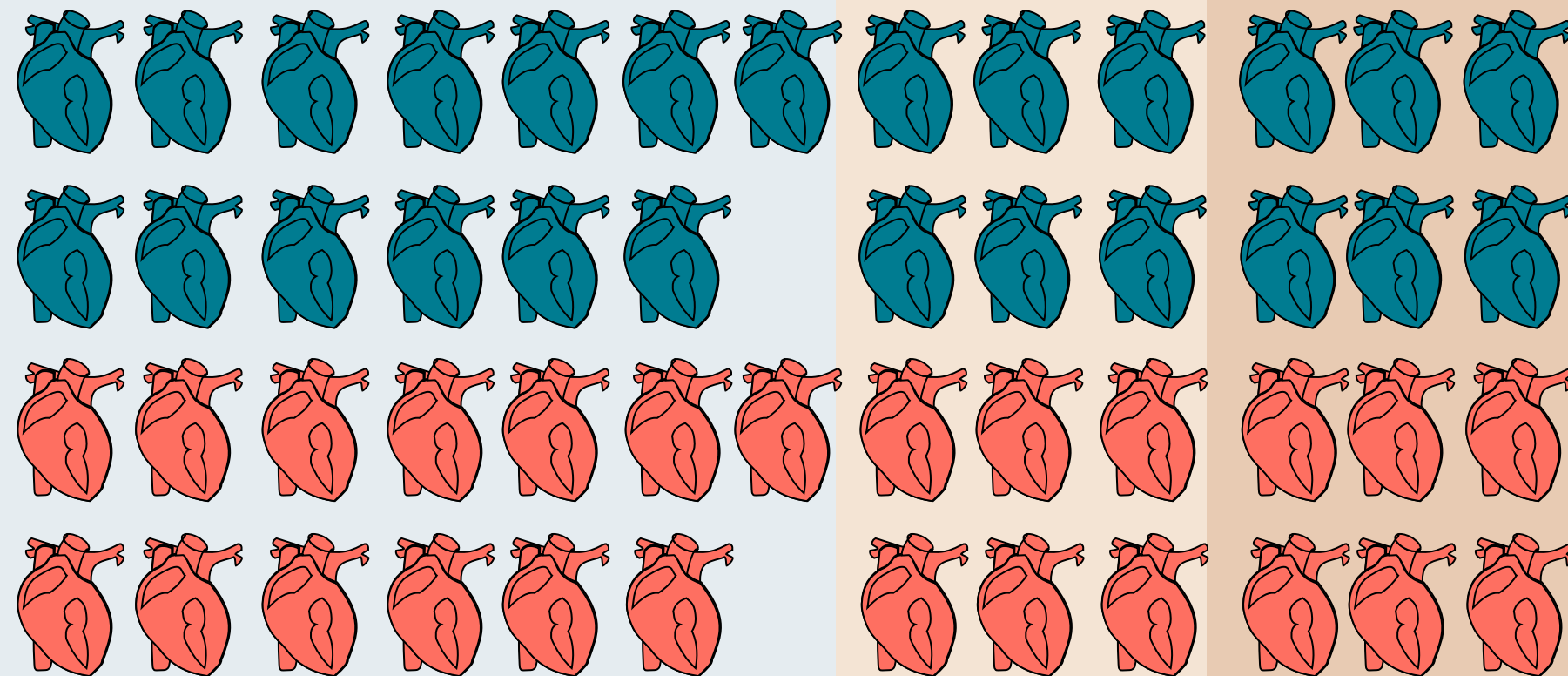
# Clinical Images

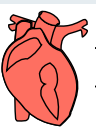
Control ( $n = 26$ )

Heart Failure ( $n = 24$ )

Narrow QRS

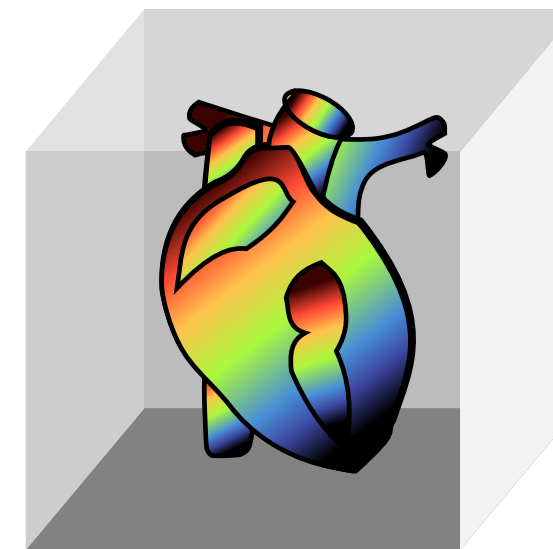
Wide QRS



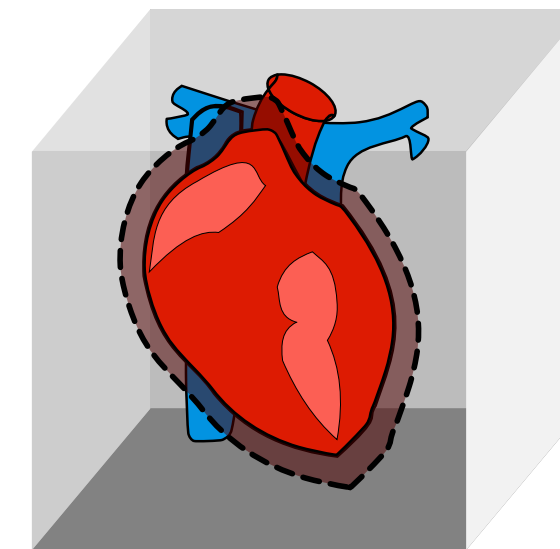
 Male  Female

# Outputs

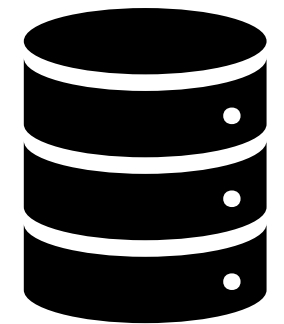
## Simulations



Electrophysiology



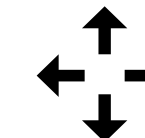
Mechanics



Sex-balanced cohort



Total Activation Times

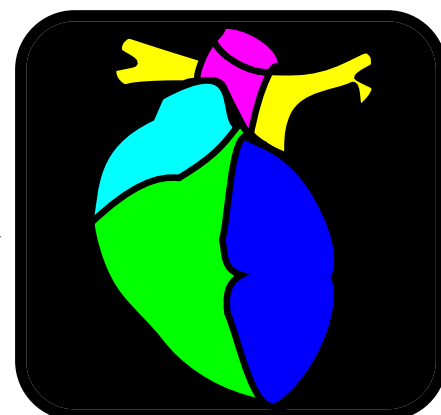


Volume  
Volume Change

## CEMRG Heartbuilder



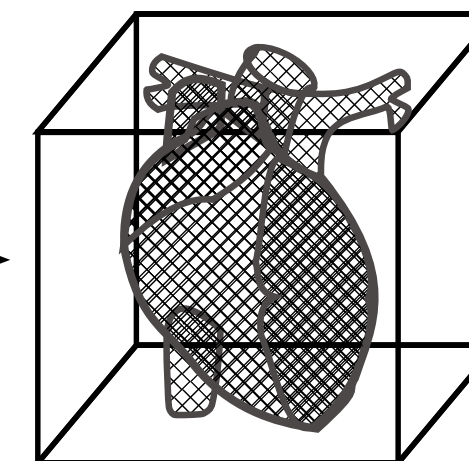
CT Scan



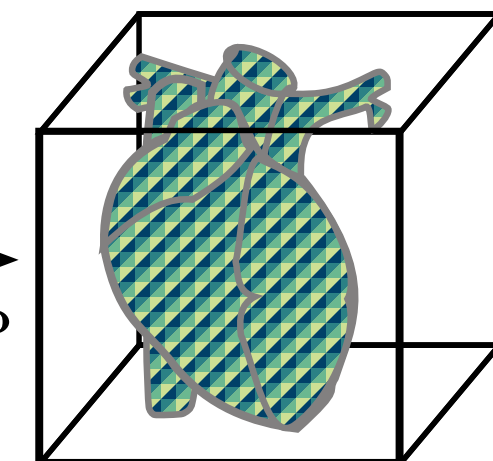
10 labels



37 labels



Volumetric Mesh



Full model

CARP